

Identification of peptide hormones of the amphipathic helix class using the helical hydrophobic moment algorithm.



Eisenberg's helical hydrophobic moment (less than μ H greater than) algorithm was applied to the analysis of the primary structure of amphipathic alpha-helical peptide hormones and an optimal method for identifying other peptides of this class determined. We quantitate and compare known amphipathic helical peptide hormones with a second group of peptides with proven nonamphipathic properties and determine the best method of distinguishing between them. The respective means of the maximum 11 residue less than μ H greater than for the amphipathic helical and control peptides were 0.46 (+/-0.07) and 0.33 (0.07) ($P = 0.004$). To better reflect the amphipathic potential of the entire peptide, the percent of 11 residue segments in each peptide above a particular less than μ H greater than was plotted vs less than μ H greater than. The resulting curves are referred to as HM-C. The mean HM-C (of the two groups) was highly significantly different such that the HM-C method was superior to others in its ability to distinguish amphipathic from nonamphipathic peptides. Several potential new members of this structural class were identified using this approach. Molecular modeling of a portion of one of these, prolactin inhibitory factor, reveals a strongly amphipathic alpha helix at residues 4-21. This computer-based method may enable rapid identification of peptides of the amphipathic alpha-helix class.